

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I Dillilokesh D 192311097 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Dillilokesh D

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. Handling Seasonal Traffic Spikes: Auto-Scaling and Load Balancing

The Problem

Seasonal sales create demand that's nowhere close to steady — traffic might sit at a normal baseline for weeks and then jump five or ten times over during a flash sale. Provisioning fixed servers for that peak means paying for idle capacity most of the year; provisioning for the average means the site falls over exactly when it matters most.

How Auto-Scaling and Load Balancing Solve It

- **Load balancing:** A load balancer sits in front of the application fleet and spreads incoming requests evenly across all available servers, so no single server gets overwhelmed while others sit idle. It also continuously health-checks each server and stops routing traffic to any that go unhealthy.
- **Auto-scaling:** An auto-scaling policy watches metrics like CPU usage, request count, or queue depth, and automatically adds new server instances the moment demand climbs — then removes them again once the spike passes, so capacity tracks demand in near real time rather than being fixed in advance.
- **Combined effect:** Together, these two mechanisms mean the platform can absorb a sudden surge of shoppers without a human manually provisioning servers in the middle of the sale.

Impact on Downtime and User Experience

Because new capacity comes online automatically as load increases, the system avoids the classic 'sale crashed the site' scenario — pages keep loading, checkout keeps working, and response times stay low even as traffic multiplies. Removing capacity after the spike also keeps the ongoing bill proportional to actual usage rather than the peak. For shoppers, this translates directly into fewer failed page loads, faster checkout, and no abandoned carts caused by timeouts.

Where Monitoring Tools Come In

- **Feeding the scaling decisions:** Monitoring dashboards track CPU, memory, request latency, and error rates in real time, which is the data auto-scaling policies actually act on.
- **Early warning:** Alerts fire before a problem becomes visible to customers, giving the operations team a chance to intervene if auto-scaling isn't keeping pace.
- **Continuous improvement:** Post-event data from monitoring tools helps the team tune scaling thresholds and capacity plans ahead of the next sale.

Real-World Example

E-Commerce Scenario:

During a Black Friday-style sale, a retailer's site sees traffic jump far beyond its normal daily average within minutes of the sale opening. A load balancer distributes the flood of requests across the web tier, while auto-scaling groups add new application servers as CPU utilization crosses a threshold. Cloud monitoring dashboards track response times and error rates throughout the event, and the extra servers are automatically terminated once traffic returns to normal after the sale — keeping the site fast during the rush and the infrastructure bill reasonable afterward.

2. Secure, Durable Cloud Backup Storage

Durability and Redundancy

Cloud storage services achieve extremely high durability by never keeping a single copy of any object. Data is automatically split and replicated across multiple physical drives, and typically across multiple data centers in different locations, so the failure of one disk, server, or even an entire facility doesn't result in data loss. Some services also apply erasure coding, which reconstructs data from partial fragments, giving durability comparable to full replication at a lower storage cost.

Protecting Sensitive Backup Data

Control	How It Protects the Data
Encryption at rest	Backup files are stored in encrypted form, so anyone who gains raw access to the underlying disks still can't read the data without the encryption key.
Encryption in transit	Data is encrypted (TLS) while moving between the organization and the cloud, preventing interception during upload or restore.
Access control (IAM/RBAC)	Only specific authenticated identities and roles can read, write, or delete backups, limiting exposure even if credentials elsewhere are compromised.

Control	How It Protects the Data
Key management	Encryption keys are managed separately from the data itself (often in a dedicated key management service), so possessing the storage doesn't mean possessing the key.
Immutability / versioning	Write-once or versioned storage prevents backups from being silently altered or deleted, which also protects against ransomware overwriting backup copies.

Cost Advantages Over Traditional Storage

- **No hardware capital expense:** No upfront cost for tape libraries, disk arrays, or a physical facility to house them.
- **Pay only for capacity consumed:** Storage is billed for what's actually used, and can shrink or grow with the organization's backup footprint.
- **Tiered storage pricing:** Rarely-accessed older backups can move automatically to cheaper archival storage tiers, cutting cost further without any manual process.
- **Reduced maintenance overhead:** The cloud provider handles hardware replacement, redundancy, and facility maintenance, removing an entire category of operational spend.

Disaster Recovery Example

Use Case:

A company keeps nightly backups of its production databases in cloud object storage, replicated automatically across two geographic regions. When a fire damages the company's primary on-premises server room, the IT team restores the most recent backup from the cloud into a new environment in a matter of hours rather than days, because the data was never dependent on the damaged facility in the first place — this is exactly the kind of scenario cross-region redundancy is built for.

3. Global App Deployment: CDNs and Edge Performance

How CDNs Improve Performance

A Content Delivery Network is a distributed network of servers positioned around the world that cache copies of an app's static content — images, videos, scripts, app assets — close to end users. Instead of every request traveling back to one origin server (possibly on another continent), the request is served from the nearest CDN node. This shortens the physical distance data has to travel, which directly reduces load times and latency, and it also takes a huge amount of repetitive load off the origin server.

The Role of Edge Servers

- **Localized delivery:** A user in Tokyo and a user in London are both served from a server physically near them rather than a single distant origin, so both get comparably fast load times.
- **Compute at the edge:** Increasingly, edge servers don't just cache content — they run lightweight compute (edge functions) to personalize content, authenticate requests, or apply business logic without a round trip to the origin.
- **Resilience:** If one edge node or region has an outage, traffic can be redirected to the next-nearest node, so the failure is contained rather than global.

How Cloud Providers Support Global Availability

- **Multi-region infrastructure:** Deploying application instances in multiple geographic regions so users are served from the closest one.
- **Global load balancing / geo-DNS:** Routing users to the nearest healthy region automatically based on their location.
- **Data replication:** Replicating data across regions so the same up-to-date content is available no matter which region serves the request.

Real-Time Application Example

Streaming / Gaming Scenario:

A mobile game studio launching worldwide deploys its game servers across several cloud regions and pushes all game assets (textures, audio, video) through a CDN. A player in Brazil connects to a nearby regional server for low-latency real-time gameplay, while their download of game assets is served from a CDN edge node in the same country rather than the studio's primary data center overseas. The result is

playable latency and fast load-in no matter where in the world the player is — which is the difference between a game feeling responsive and feeling unplayable.

4. Compliance and Security for a Financial Institution in the Cloud

How Cloud Providers Support Regulatory Compliance

Major cloud providers maintain certifications against standards relevant to finance — PCI-DSS for payment data, SOC 2 for security controls, ISO 27001 for information security management, and region-specific regulations. They publish compliance reports and audit artifacts that the institution can use as evidence during its own regulatory audits, and they offer purpose-built compliance dashboards that continuously check configurations against required controls.

Logging, Monitoring, and Encryption

Security Element	Role in a Financial Context
Logging	Every access to sensitive systems and data is recorded (who, when, what action), creating the audit trail regulators require and letting the institution reconstruct events during an investigation.
Monitoring	Real-time analysis of logs and system behavior flags unusual activity — an account accessing data outside normal patterns, for instance — often before it becomes a breach.
Encryption	Financial data is encrypted at rest and in transit, so even if storage or network traffic is intercepted, the data itself remains unreadable without the key.

Meeting Data Sovereignty Requirements

- **Region selection:** Choosing cloud regions physically located within the country or jurisdiction whose laws require the data to stay there.
- **Data residency controls:** Using data residency controls to guarantee that specific data types never replicate outside approved geographic boundaries.

- **Customer-managed encryption keys:** Keeping encryption keys under the institution's own control (or a locally-based key management service) rather than solely the provider's, adding another layer of jurisdictional control.

Example Involving Financial Transactions

Scenario:

A bank processing card payments stores transaction records in a cloud region located in the customer's home country to satisfy data residency law. Every transaction is encrypted in transit using TLS and at rest using keys the bank manages itself. All access to the transaction database is logged and continuously monitored, and an automated alert fires when an account attempts to query an unusually large volume of records outside business hours — flagging a potential insider threat before any data leaves the system.

5. Containerization and Orchestration for Application Deployment

Containers vs Virtual Machines

Aspect	Virtual Machines	Containers
What's virtualized	Full hardware, including its own guest OS per VM	The OS kernel is shared; only the application and its dependencies are packaged
Startup time	Minutes (booting a full OS)	Seconds (no OS boot required)
Resource footprint	Heavier — each VM carries a full OS	Lightweight — only the app and its libraries
Density per host	Fewer VMs fit per physical machine	Many more containers can run on the same hardware
Isolation strength	Very strong (separate kernels)	Slightly weaker (shared kernel) but strong enough for most workloads

How Orchestration Simplifies Deployment and Scaling

- **Automated scheduling and self-healing:** Kubernetes automatically restarts failed containers, spreads them across available nodes, and can move them if a node goes down — all without manual intervention.
- **Horizontal scaling:** The number of running container instances (pods) scales up or down automatically based on load, the same principle as VM auto-scaling but at a faster, lighter-weight level.
- **Rolling updates:** New versions of an application roll out gradually across containers, with automatic rollback if something breaks — deployments that used to be risky, manual events become routine.
- **Service discovery:** Built-in service discovery and internal DNS let containers find and talk to each other automatically as they're created, destroyed, and moved around.

Portability Across Cloud Environments

A container packages the application together with everything it needs to run, so the same container image behaves identically whether it's running on a developer's laptop, an on-premises server, or any major cloud provider's Kubernetes service. This portability is what makes multi-cloud and hybrid-cloud strategies practical — a team isn't rewriting deployment logic every time they move workloads between AWS, Azure, or Google Cloud, because Kubernetes provides the same interface everywhere.